

Algorithms Problem Solving

Level 2

Encrypt Decrypt Text

The background of the slide is a photograph of a modern building's interior. It features a large, curved glass ceiling that allows natural light to filter through, creating a pattern of light and shadow on the floor. In the lower portion of the image, the silhouettes of several people are visible, some standing and others sitting, suggesting a collaborative workspace or a public area. The overall color palette is muted, with greys, blues, and the warm tones of the light coming through the glass.

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Definition

- ◆ A program that reads a text, encrypts it by shifting each character forward by a specific key, then decrypts it by shifting each character backward by the same key.
- ◆ In this example, the encryption key is `2`.

Explanation

- ◆ ``ReadText()`` uses ``getline()`` to read the complete text, including spaces.
- ◆ ``EncryptText()`` loops through every character in the text.
- ◆ Each character is converted to its integer character code using ``(int)text[i]``.
- ◆ The encryption key is added to the character code.
- ◆ The result is converted back to a ``char`` and stored in the same position.
- ◆ ``DecryptText()`` performs the opposite operation by subtracting the same encryption key.
- ◆ Because the same key is used in both operations, the original text can be restored.

Steps in main()

- 1 Read the text using ``getline()``.
- 2 Set the encryption key to ``2``.
- 3 Print the original text.
- 4 Loop through every character in the text.
- 5 Add ``2`` to each character's code to encrypt it.
- 6 Store the encrypted characters in the string.
- 7 Print the encrypted text.
- 8 Loop through the encrypted text again.
- 9 Subtract ``2`` from each character's code to decrypt it.
- 10 Print the decrypted text.

💡 Example

◆ Input:

Sara

◆ Encryption key:

2

◆ Each character is shifted by `2` :

`S → U`

`a → c`

`r → t`

`a → c`

◆ Therefore:

Sara → Uctc

➡ Result

◆ The program prints:

Text Before encryption: Sara

Text After encryption: Uctc

Text After decryption: Sara

◆ The encrypted text is `Uctc`.

◆ After decryption, the original text `Sara` is restored.

✚ Note

- ◆ `text[i]` accesses the character at position `i`.
- ◆ `text.length()` returns the number of characters in the string.
- ◆ The loop:
`for (int i = 0; i < text.length(); i++)`
processes every character from the first character to the last.
- ◆ Encryption adds the key:
`character + key`
- ◆ Decryption subtracts the key:
`character - key`
- ◆ The encryption and decryption operations are therefore opposites.

⊘ Warning

- ◆ This is a simple character-shifting technique for learning purposes, not secure modern encryption.
- ◆ A fixed key such as `2` can be easily discovered and reversed.
- ◆ Character shifting can also produce unexpected characters when the character code moves outside the expected character range.

Comparison

- Encryption → adds the key to each character.
- Decryption → subtracts the key from each character.
- Original text → `Sara`
- Encrypted text → `Uctc`
- Decrypted text → `Sara`

✦ Summary

- ✓ ``getline()`` reads the complete text.
- ✓ ``text[i]`` accesses each character.
- ✓ ``(int)text[i]`` treats the character as a numeric code.
- ✓ Encryption adds the key to every character.
- ✓ Decryption subtracts the same key.
- ✓ The same key must be used to correctly decrypt the text.
- ✓ ``Sara`` becomes ``Uctc`` with key ``2``.
- ✓ Decrypting ``Uctc`` with key ``2`` returns ``Sara``.

Thank you 😊